

SHERRY WILSON

Senior Python & Django Backend Developer | Software Engineer

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LinkedIn: [linkedin.com/in/sherry-wilson-python-dev](https://www.linkedin.com/in/sherry-wilson-python-dev) | Visa: Graduate Route (valid until April 2027; will require sponsorship thereafter)

PROFESSIONAL SUMMARY

Results-driven Senior Python Backend Developer and Software Engineer with 4+ years of professional experience designing, building, and deploying scalable REST APIs, GraphQL services, and cloud-native backend systems. Expert in Python, Django, Django REST Framework (DRF), Celery, PostgreSQL, Docker, and AWS. Proven track record of leading backend engineering teams, optimising system performance, and delivering enterprise-grade platforms serving tens of thousands of concurrent users. Achieved 30% throughput improvement, 25% latency reduction, and 15% revenue uplift for production systems. Skilled in microservices architecture, CI/CD pipelines, asynchronous task processing, event-driven systems, and agile software development. MSc Artificial Intelligence & Its Applications, University of Essex.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, HTML, CSS, Bash

Frameworks & Libraries: Django, Django REST Framework (DRF), FastAPI, Flask, Celery, LangChain, Pandas, NumPy, Matplotlib, GraphQL (Graphene)

Databases: PostgreSQL, Redis, ChromaDB, MySQL, SQLite

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Docker, Docker Compose, Nginx, Linux, Git, GitHub Actions, CI/CD, Load Balancing, Terraform (basic)

Messaging & Streaming: Apache Kafka, Redis Pub/Sub, Celery, Flower, RabbitMQ, WebSockets, HLS, FFmpeg

AI & Machine Learning: LangChain, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Google Gemini, OpenAI API, NLP, Data Analysis, Feature Engineering

Architecture & Methodologies: Microservices, Event-Driven Architecture, REST API Design, GraphQL Schema Design, Test-Driven Development (TDD), Agile, Scrum, System Design, Domain-Driven Design (DDD)

PROFESSIONAL EXPERIENCE

Senior Python Backend Developer / Team Lead

Aug 2021 – Jul 2024

Deepsense Digital Solutions — Chennai, India

- Led a team of 3 backend engineers delivering production-grade Python/Django systems for enterprise clients: ITC (Classmate), Fabelle, Greencell Mobility (Neugo), and Nippon Paints.
- Architected and developed scalable RESTful APIs and backend services using Django, Django REST Framework, and PostgreSQL, serving 50,000+ active users across multiple platforms.
- Designed and implemented a GraphQL API layer (Graphene-Django), reducing frontend-server data transfer by 30% and improving page load performance.
- Built distributed asynchronous task processing pipelines using Celery, Redis, and Flower, improving system throughput by 30% and enabling real-time task monitoring and observability.
- Performed PostgreSQL query optimisation, indexing, and connection pooling, achieving stable sub-200ms response times under peak traffic loads.

- Deployed and managed containerised microservices using Docker, AWS EC2, Nginx reverse proxy, and load-balanced architecture achieving 99.9% uptime SLA.
- Designed Kafka-based event-driven microservices communication, reducing inter-service latency by 25% and improving system decoupling.
- Built adaptive bitrate HLS video streaming pipelines with FFmpeg and AWS S3, improving media delivery speed by 30% for eLearning platforms at scale.
- Delivered backend infrastructure improvements that directly contributed to a 15% client revenue uplift through platform scalability, reliability, and performance gains.
- Conducted code reviews, mentored junior developers, and established team coding standards and CI/CD best practices.

Python LLM Developer

Sep 2025 – Oct 2025

AquSag Technologies — Remote

- Developed evaluation benchmarks for Large Language Models (LLMs) by analysing open-source Python repositories and triaging high-impact GitHub issues for automated code-generation testing.
- Assessed unit test coverage and code quality metrics to measure LLM performance in real-world bug-fixing and code generation tasks.
- Collaborated with AI researchers to design repository benchmarks for rigorous LLM engineering capability assessment.
- Mentored junior developers on Python best practices, testing methodologies, and delivery standards.

AI/ML Intern — Data Engineering

Jul 2025 – Aug 2025

Victoria Solutions — London, UK

- Performed end-to-end data analysis, cleaning, and transformation with Python and Pandas for AI model training datasets, improving data quality metrics by 40%.
- Developed automated reporting dashboards using Matplotlib to communicate model performance and dataset insights to stakeholders.
- Engineered and structured large-scale datasets for ML pipelines, accelerating model training and feature engineering workflows.

Freelance Python Backend Developer

Jun 2020 – Jul 2021

Self-employed — Kerala, India

- Designed and delivered full-stack backend solutions using Django, Flask, and FastAPI for clients requiring secure, scalable REST API architectures.
- Implemented automated background processing workflows with Celery and Redis, reducing manual operations by 60%.
- Containerised applications with Docker and deployed across cloud environments with CI/CD pipelines.
- Integrated machine learning models into production systems via secure Python REST APIs with authentication and rate limiting.

KEY ACHIEVEMENTS

- Improved backend system throughput by 30% through Celery/Redis asynchronous processing across enterprise production platforms.
- Reduced inter-service latency by 25% via Kafka-based event-driven microservices architecture redesign.
- Delivered 15% revenue uplift for enterprise clients through scalable Python/Django backend architecture and reliability improvements.
- Built HLS video streaming infrastructure serving thousands of concurrent learners with 30% faster delivery.

- Led backend team delivery for 4 enterprise client platforms simultaneously within deadline and budget constraints.
- Reduced frontend-server data transfer by 30% through optimised GraphQL schema design and query batching.

PROJECTS

Document Assistant API (RAG): Production-ready FastAPI service using LangChain, ChromaDB, and Google Gemini for intelligent document Q&A with citation-grounded responses. Tech: Python, FastAPI, LangChain, ChromaDB, Gemini, Docker, REST API.

HLS Video Streaming Platform: Adaptive bitrate streaming pipelines with FFmpeg, AWS S3, and Nginx; achieved 30% faster media delivery at scale. Tech: Python, FFmpeg, AWS S3, Nginx, Docker, HLS, Redis.

Multi-Tenant E-Commerce Backend: Django DRF + GraphQL backend with Celery, Kafka, PostgreSQL, and Docker serving thousands of concurrent users. Tech: Django, DRF, GraphQL, Celery, Redis, Kafka, PostgreSQL, Docker.

ML Data Pipeline & Dashboard: Automated data ingestion, cleaning, feature engineering, and visual reporting for AI model training workflows. Tech: Python, Pandas, NumPy, Matplotlib, PostgreSQL, Celery.

EDUCATION

Master of Science (MSc) — Artificial Intelligence & Its Applications Jan 2024 – Jan 2025
University of Essex, Colchester, UK

Bachelor of Computer Applications (BCA) — Computer Programming 2018 – 2021
University of Kerala, India

CERTIFICATIONS

- FastAPI — The Complete Course 2025 (Beginner + Advanced)
- Advanced Django Development
- Full UK Driving Licence

ADDITIONAL EXPERIENCE

Team Leader Jan 2026 – Present
Welshwood Manor — Colchester, UK

Senior Healthcare Assistant Jul 2024 – Jan 2026
Care Circle — Colchester, UK

UK-based leadership roles demonstrating team management, stakeholder communication, and operational delivery while actively building backend and AI projects for the technology sector.